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%% Verlorenvlei Groundwater Isotope Analysis

clear; clc; close all;

%% Load + prepare data
dt = readtable('Verlorenvlei_Groundwater_isotopes.xlsx', ...
    'VariableNamingRule','preserve');

dt.Date = datetime(dt.Date);

% Define sampling groups
grp = strings(size(dt.Date));
grp(month(dt.Date)==3 & year(dt.Date)==2023) = "Mar 2023";
grp(month(dt.Date)==6 & year(dt.Date)==2023) = "Jun 2023";
grp(month(dt.Date)==11 & year(dt.Date)==2023) = "Nov 2023";
grp(month(dt.Date)==1 & year(dt.Date)==2024) = "Jan 2024";

% Extract
d2H = str2double(string(dt{:, 'δ2H (‰)'}));
d18O = str2double(string(dt{:, 'δ18O (‰)'}));
dExcess = str2double(string(dt{:, 'd-excess (‰)'}));
H3 = str2double(string(dt{:, 'H3 (TU)'}));

% Remove NaNs
valid = ~isnan(d2H) & ~isnan(d18O) & ~isnan(dExcess) & ~isnan(H3);
[d2H,d18O,dExcess,H3,grp] =
deal(d2H(valid),d18O(valid),dExcess(valid),H3(valid),grp(valid));

% Labels + colors
lbl = ["Mar 2023","Jun 2023","Nov 2023","Jan 2024"];
clrs = [0 0.4 1; 0 0.7 0.7; 0 0.7 0.2; 1 0.2 0.2];

% GMWL
x = linspace(min(d18O)-1, max(d18O)+1, 100);
GMWL = 8*x + 10;

%% ===== PLOTS =====
figure;

% ----- Subplot 1 -----
subplot(2,2,1); hold on;
for i = 1:4
    scatter(d18O(grp==lbl(i)), d2H(grp==lbl(i)), 20, ...
        'filled', 'MarkerFaceColor', clrs(i,:));
end
hG = plot(x, GMWL, 'k:', 'LineWidth', 2);

xlabel('\delta^{18}O (‰)'); ylabel('\delta^{2}H (‰)');
title('δ^{18}O vs δ^{2}H'); grid on; box on;

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lgd1 = legend([lbl, "GMWL"], 'Location','best');
lgd1.ItemTokenSize = [5 10];
hold off;

% ----- Subplot 2 -----
subplot(2,2,2); hold on;
for i = 1:4
    scatter(d180(grp==lbl(i)), dExcess(grp==lbl(i)), 20, ...
        'filled', 'MarkerFaceColor', clrs(i,:));
end
yline(10,'k--','LineWidth',1.5);

xlabel('\delta^{18}O (‰)'); ylabel('d-excess (‰)');
title('d-excess vs  $\delta^{18}O$ '); grid on; box on;

lgd2 = legend(lbl, 'Location','best');
lgd2.ItemTokenSize = [5 10];
hold off;

% ----- Subplot 3 -----
subplot(2,1,2); hold on;
scatter(d180, d2H, 20, H3, 'filled');
plot(x, GMWL, 'k:', 'LineWidth', 2);

colorbar; colormap jet;
xlabel('\delta^{18}O (‰)'); ylabel('\delta^{2}H (‰)');
title('δ^{18}O vs δ^{2}H (Tritium)');

grid on; box on; hold off;

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